

Evidence desk

Soil: what is in the mix?

Name _____ Date _____

Constructed teaching examples. Cut cards or point to them. Use these even if no soil is available.

A • Pale grain

Under a hand lens: a hard, gritty quartz grain. Quartz is a mineral found in rocks.

C • Tiny stem

A short woody piece with visible fibres. It came from a plant stem.

E • Bubbles

When water is poured onto this soil, bubbles rise from spaces between particles.

B • Leaf piece

A soft brown fragment with a clear leaf vein. It came from a fallen leaf.

D • Dark grain

A dark speck. Its colour is visible, but its material is not identified.

F • Water

A damp sample leaves a water mark on absorbent paper. No liquid was added during this observation.

W1. Sort A, B and C. Give one evidence clue.

W2. Which of D-F needs more evidence? Why?

Route 1 • Match and explain

Choose this route for supported recording. An adult may read or scribe your words.

S1. A is a quartz grain. Choose its group.

A. Mineral material

B. Organic matter

S2. B is a fallen-leaf piece. Choose its group.

A. Mineral material

B. Organic matter

S3. Use a clue: I put C with organic matter because ____.

S4. Soil is a mixture of ____ and ____. Word bank: mineral material / organic matter.

S5. Draw or name a soil pore. What might be in it?

Route 2 • Write an evidence record

Use cards A-F. Record only what the evidence supports.

Task	Card	Component or uncertainty	Evidence clue
T1	A		
T1	B		
T1	C		
T1	D		
T1	E		
T1	F		

T2. Explain why soil is more than broken rock.

T3. A classmate says “D is organic because it is dark.” What would you say?

Route 3 • Improve the model

A designer draws soil as tightly packed grey stones, with no gaps.

X1. Sketch a better model. Label mineral material, organic matter and one pore.

X2. Add air or water. Explain where you put it.

X3. Why can the dark-grain card not decide its material?

X4. What can your model help explain? What does it leave out?

Exit and reflection

Complete the three exit checks. You may point, draw, write or tell an adult.

E1. Name or draw one mineral example from today.

E2. Name or draw one organic example from today.

E3. What can fill a soil pore?

R1. One idea I changed after hearing evidence was ____.

R2. Next time, I would like to observe or ask ____.
